25901613 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION APRIL 2026

SECOND SEMESTER

Multi-Disciplinary Courses (MDC) - MG2MDCBTG100 - TOOLS AND TECHNIQUES IN BIOTECHNOLOGY

(2024 ADMISSION ONWARDS)

Duration: 1 Hours **Maximum Marks: 35**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

One Word Answer

Answer any **10** questions

Each question carries **1** marks

1. Which enzyme prevents DNA supercoiling? [U] / [CO1]
2. How do restriction exonucleases differ from restriction endonucleases? [K] / [CO2]
3. Define real time PCR. [K] / [CO1, CO4]
4. Define Electrophoresis. [K] / [CO1]
5. Name two methods of gene cloning. [U] / [CO2]
6. Comment on the purpose of NCBI. [U] / [CO3]
7. Cite the methods of site directed mutagenesis. [U] / [CO2]
8. Describe annealing in PCR. [U] / [CO1, CO4]
9. What are the steps involved in expressing a protein in a host cell by gene cloning? [U] / [CO2]

10. Mention the primary focus of bioinformatics? [K] / [CO3]

11. What factors influence the migration rate of proteins in SDS-PAGE? [K] / [CO1]

12. Identify the anionic stain used in SDS-PAGE. [K] / [CO4]

[1x10 = 10]

Part B

Short Essay

Answer any **3** questions

Each question carries **5** marks

13. Discuss the limitations of both AGE and PAGE in electrophoretic separation. [K] / [CO1]

14. Outline the main stages in Central Dogma? [K] / [CO1]

15. What is the role of t-RNA in gene expression? [U] / [CO2]

16. Describe the process of translation from initiation to termination. [U] / [CO1]

17. Examine the principle of mass spectrometry. [K] / [CO4]

[5x3 = 15]

Part C

Essay

Answer any **1** questions

Each question carries **10** marks

18. Describe the organization of DNA. [K] / [CO1]

19. Describe the differences between bacteriophage vectors and cosmid vectors. What are their advantages and limitations? [U] / [CO2]

[10x1 = 10]